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RESEARCH ARTICLE

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A retrospective study on prevalence, characteristics, and outcomes of transcribing error: analysis of data from the National Medication Error Reporting System in Malaysia

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ABSTRACT

Background: Medication errors (MEs) represent avoidable harm that occurs within the healthcare system. While most MEs do not pose significant safety threats, some can result in severe harm, disability, or even death for patients. We aimed to describe the prevalence, characteristics, and outcomes of MEs due to transcribing errors.

Methods: Retrospective analysis of data from the Medication Error Reporting System (MERS) in Malaysia. All ME reports submitted to MERS were verified and classified by trained pharmacists from the Ministry of Health Malaysia. Transcribing errors were identified from all ME reports submitted to MERS that were classified as 'data entry error'. All data from public health facilities (hospitals and clinics) submitted in 2018–2022 were included in the analysis. Descriptive analyses were conducted on frequencies and characteristics of the transcribing errors.

Results: From January 2018 to December 2022, 265,194 MEs were reported to the MERS, of which 12,049 (4.5%) were transcribing errors. The large proportion of transcribing error was related to dose error, i.e. incorrect dose (48.6%) and incorrect frequency (19.9%), followed by drug error (incorrect drug, 13.3% and incorrect patient, 6.3%). The majority of the errors do not reach patients (94.8%). The most possible error causes and contributing factors were work and environment followed by staff factors, task and technology, and medication-related.

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Conclusion: There was an increase in voluntary reporting over the years, reflecting a growing culture of reporting among healthcare professionals. These findings offer valuable insights into medication errors caused by transcribing processes in Malaysia, which can inform efforts to improve pharmacy practices, design appropriate interventions, and reduce harm to patients.

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KEYWORDS Medication error; pharmacy; transcribing error

Background

The term 'medication error' (ME) refers to any preventable event that may cause or lead to inappropriate medication use or patient harm while the medication is in the control of healthcare professionals, patients, or consumers (National Coordinating Council for Medication Error Reporting and Prevention, n.d.). MEs represent avoidable harm within the global healthcare system, with an estimated annual cost of 42 billion US dollars worldwide (World Health Organization, n.d.). The incidence of MEs varies as it depends on operational definitions, healthcare settings, and the population studied. A systematic review found that MEs ranged between 23% and 92% of prescribed drugs in outpatient and ambulatory settings (Naseralallah et al., 2023). Another study reported that MEs affected approximately 36.3% of patients in the emergency department, with 42.6% of the errors classified as potentially harmful (Nguyen et al., 2024). While many MEs do not cause major safety threats to patients, the most serious consequences occur when the ME causes severe harm, disability, and death to the patient (Alrabadi et al., 2021).

Reporting MEs is fundamental for error detection and prevention. Thus having an effective ME reporting system is of paramount importance to encourage and enhance the ability of healthcare providers to report any incidence of ME encountered (Afaya et al., 2021; Holmström et al., 2015). In Malaysia, the Pharmaceutical Services Programme, Ministry of Health, established a national-level reporting, the Medication Error Reporting System (MERS) in 2009 (Pharmaceutical Services Programme. Ministry of Health Malaysia, 2019). Reporting MEs through the MERS is a strategy to enhance medication safety, aligning with the Malaysian Patient Safety Goals (MPSG) 2.0 – Goal 3.8 (Patient Safety Council Malaysia, Medical Care Quality Section, & Medical Development Division, 2021). This national reporting system gathers data through voluntary reporting from healthcare professionals on error incidents associated with medication use. Although not mandatory, all healthcare professionals from both public and private sectors are strongly encouraged to report cases of MEs which can be submitted via online or offline platform. Since the introduction of the system, the number of reports on medication

errors has increased and reached approximately 60,000 reports annually. In addition, several initiatives have also been introduced by the Ministry of Health to mitigate MEs, including training modules for healthcare providers, patient education campaigns, and digitalisation of healthcare (Ismail et al., 2015; Ministry of Health Malaysia, 2020; Pharmaceutical Services Division & Ministry of Health Malaysia, 2012; Pharmaceutical Services Programme. Ministry of Health Malaysia, n.d.).

MEs can occur at various stages of the medication use process, such as prescribing, transcribing, preparing, dispensing, and administering. Prescribing and transcribing stages are common sources of MEs (Alshammari et al., 2022; Chalasani et al., 2018; Salmasi et al., 2015; van Doormaal et al., 2009). The digitalisation of healthcare systems such as the implementation of electronic prescribing and medication dispensing systems could reduce MEs and improve patient safety (Gates et al., 2021). However, errors may still occur especially in facilities that were only partially digitalised which may still require the transcribing of prescriptions from paper forms into an electronic system before dispensing the medicines. Concerns were raised regarding such transcribing errors as it can be a nontrivial source of error (Lindén-Lahti et al., 2022; Shawahna et al., 2019).

Given the availability of the MERS database, analysis of reporting data can further contribute to the improvement of patient safety and the health system since there remains a lack of understanding regarding the current situation of MEs in Malaysia. Moreover, limited studies are available that examined MEs related to transcribing errors. Samsiah et al. previously analysed 17,357 MEs reported to MERS between 2009 and 2012 in Malaysia; however, it was focused on the quantification of MEs by stages of the medication use process and did not provide details on specific error types (Samsiah et al., 2016). Furthermore, they used data from an earlier period when the reporting system was just established and the computerisation of the healthcare system was still in its infancy. Therefore, our current study was carried out with the primary objective of estimating the prevalence of MEs due to transcribing errors that occurred in healthcare facilities in Malaysia. Our secondary objective was to describe the characteristics and outcomes of the transcribing errors.

Methods

Study design

This was a cross-sectional, retrospective study of medication error reports submitted to the MERS from January 2018 to December 2022.

Setting

Healthcare system in Malaysia consists of two parallel sectors: government-funded public health sectors (public) and fee-for-service private sector. Public health sector is mainly provided by the Ministry of Health (MOH) which operates widespread network of hospitals and clinics across Malaysia. Each MOH facilities have its pharmacy department manned by pharmacists and pharmacy assistants and responsible for pharmaceutical care and supply of medicines for patients at the respective facilities. Over the past decade, several types of electronic health information systems were in use across MOH facilities. The system can be classified as (i) full-based, (ii) pharmacy-based, and (iii) manual (paper-based). Facilities with a full-based system are paperless facilities with the integration of a fully computerised system across administrative, clinical, and financial workflow. For pharmacy-based, only the pharmacy system is computerised where the process of ordering medication is paper-based (manual prescriptions) and the process of dispensing the medication is computerised. Facilities with manual systems have yet to implement health information systems and the process workflow is paper-based.

Data source

Data extracted from the MERS database was provided by the Pharmaceutical Service Programme, Ministry of Health (MOH) Malaysia. ME reports in Malaysia can be submitted to MERS by healthcare professionals using a standardised reporting form following a structured process and according to specified guideline (Pharmaceutical Services Programme. Ministry of Health Malaysia, 2019). All ME reports were reviewed and verified by trained officers of MOH in several stages. Details of the submission of ME reports to MERS and data verification process are outlined in the Supplemental Material. The ME report form collects information on time of event, location of the event, demographic of patients (age, gender), stages of the medication process and type of error, error outcome, possible error cause and contributing factor, and product details (drug name, dosage form, dose, frequency, route, strength). The ME report was anonymised and did not collect information on personal information and clinical details.

For the current study, only ME data from public health facilities (hospitals and primary care clinics operated by MOH) were included in the analysis. Data on total number of prescriptions processed at each healthcare facility over the study period was provided by the Pharmaceutical Services Programme, MOH, which collected the information from the facilities as performance monitoring indicators.

Definitions

MEs were classified based on the process in which it occurs: prescribing, data entry system, labelling, filling, counter dispensing, and administration. Data entry, labelling, filling, and counter dispensing are four key sub-processes originating from the dispensing process by the pharmacy. In this study, we identified all ME reports classified as data entry system errors. These were errors that occurred in the process of interpreting, verifying, and transferring information of medication orders and were subsequently coded as transcribing errors. MEs due to data entry that occurred during the process of ordering medication (prescribing) were not included.

The following information was extracted from the database for analysis: description of MEs, error type, error outcome, error causes and contributing factors, medication involved in error, reporter information, and facilities. Medication was categorised according to the World Health Organization (WHO) Anatomical Therapeutic Chemical (ATC) classification. MEs are classified according to the severity of outcome or degree of harm, from Category A (no error – the event can cause error) to Category I (error occurred – event may contributed to or result in patient's death). Definition of types of medication errors is provided in Supplemental Material Table S1.

Statistical analysis

Prevalence of transcribing errors was calculated using the number of transcribing errors divided by the total number of prescriptions. Percentage of transcribing errors was computed using the total number of transcribing errors over the total number of all ME reports to describe the share of transcribing errors across all ME subtypes. Descriptive statistics were used to present the characteristics of errors in frequencies and percentages. Comparisons between groups were made using the Chi-square test or Fisher's exact test. A p-value <0.05 is considered to be statistically significant. Analysis was conducted using STATA SE 15.1 (StataCorp LLC, College Station, TX, USA).

Ethical approval

This study was approved by the Medical Research and Ethics Committee, Ministry of Health Malaysia (NMRR ID-24-00304-OBO) with a waiver of informed consent due to the use of secondary, de-identified data for analysis.

Results

From January 2018 to December 2022, 265,194 MEs were reported to the MERS. In the medication use process, prescribing, dispensing, and

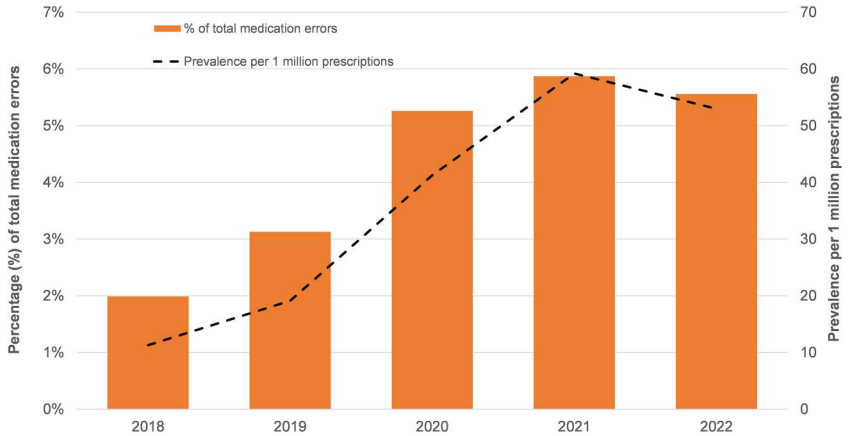


Figure 1. Percentage of medication errors due to transcribing errors and prevalence per total number of prescriptions between 2018 and 2022.

Note: Percentage was calculated using the total number of medication error reports as denominator and prevalence was calculated using the total number of prescriptions as denominator.

administration contributed to medication errors at rates of 70.5%, 28.4%, and 1.0% respectively. Among errors originating from the dispensing process by the pharmacy, four key sub-processes were identified: data entry (4.5%), labelling (2.1%), filling (21.3%), and counter dispensing (0.4%). Breakdown of all medication errors during 2018–2022 by medication use process is described in Supplemental Material Table S2. Of these, a total of 12,049 reports (4.5%) were identified as transcribing errors, which corresponded to 34.9 events per 1 million prescriptions processed over this period. Figure 1 describes the annual percentage and prevalence of transcribing errors over the 5-year study period, which showed an upward trend with an increment from 11.3 per million prescriptions ($n = 879$, 2.0% of all MEs) in 2018 to 53.0 per million prescriptions ($n = 3544$, 5.6% of all MEs) in 2022.

Table 1 provides an overview of the 12,049 transcribing errors included in the analysis. Hospitals contributed 47.4% of the errors while the rest were from primary care clinics (52.6%). A large majority of the errors were reported by facilities with pharmacy-based information systems (97.5%). These errors mostly occurred in pharmacy ($n = 11,834$, 98.2%); of these, outpatient pharmacy had the highest frequency (88.6%) while inpatient and satellite pharmacy accounted for only 8.2% and 1.7%, respectively. The majority of the errors were detected by pharmacists (92.2%) and occurred during normal working hours (94.0%).

Table 2 summarises the types of transcribing errors and error outcomes or degree of harm. The largest share was related to dose error, i.e. incorrect dose (48.6%) and incorrect frequency (19.9%). Incorrect drug (i.e. wrong drug, right patient) accounted for 13.3% while there were 6.3% instances of incorrect

Table 1. Characteristics of medication errors due to transcribing errors, 2018–2022.

	Number of events, *n*	Percentage, %
N	12,049	
Year		
2018	879	7.3%
2019	1478	12.3%
2020	2644	21.9%
2021	3504	29.1%
2022	3544	29.4%
Facility – type		
Hospital	5706	47.4%
Clinic	6343	52.6%
Facility – status of health information system[a]		
Full-based	277	2.3%
Pharmacy-based	11,758	97.5%
Manual	61	0.5%
Not specified	14	0.1%
Location of event		
Pharmacy	11,834	98.2%
Ward	40	0.3%
Clinic	152	1.3%
Accident and Emergency	11	0.1%
Others	12	0.1%
Time of event		
Office hoursb	11,327	94.0%
Outside office hours	722	6.0%
Error detected by		
Pharmacist	11,110	92.2%
Physician	60	0.5%
Other healthcare Professionals	769	6.4%
Consumer/caregiver	109	0.9%
Others	1	<0.1%
Error reached patient		
Yes	534	4.4%
No	11,515	95.6%

[a]Full-based = Fully electronic hospital system; pharmacy-based = pharmacy stand-alone system.
[b]Office hours refer to time between 0730 and 1730 on weekdays.

patient. There were 448 cases (3.7%) of omission errors in which prescribed or ordered medication (s) were incorrectly left out during transcribing. Dosage form errors accounted for 4.8%, which include incorrect formulation (e.g. metformin instead of metformin sustained release) and incorrect dosage form (e.g. tablet instead of intravenous). Of note, the majority of errors did not reach patient (95.5%). Meanwhile, 434 (3.6%) were classified as Category C (error occurred that reached patient, but did not cause harm), 13 errors (0.1%) were classified as Category E (error caused temporary harm and required treatment/intervention), and 3 errors (0.02%) in Category F (error caused temporary harm and required initial/prolonged hospitalisation).

Medication involved in MEs is described in Figure 2 according to ATC grouping at level 1. The most frequently reported medications were drugs for the cardiovascular system (29.9%) and drugs for the alimentary tract and metabolism (26.9%). Breakdowns by therapeutic subgroups (ATC level 2) are provided in Table S3. Drugs used in diabetes, lipid-modifying agents,

Table 2. Error type and outcome (degree of harm).

	Total *n* (%)
Total errors, *N*	**12,049**
Error type	
Dose error	
Incorrect dose	5889 (48.6)
Incorrect frequency	2410 (19.9)
Drug error	
Incorrect drug	1608 (13.3)
Incorrect patient	765 (6.3)
Dosage form error	
Incorrect dosage form	587 (4.8)
Omission error	
Omission of drug	448 (3.7)
Labelling error	
Incorrect instruction	64 (0.5)
Other transcribing error	
Incorrect time	19 (0.2)
Incorrect preparation	5 (<0.1)
Incorrect reporting	3 (<0.1)
Others	251 (2.1)
Error outcome	
A: Potential error, events have potential to cause incident	81 (0.7)
B: Actual error – did not reach patient (near miss)	11,427 (94.8)
C: Actual error – caused no harm	432 (3.6)
D: Additional monitoring required – caused no harm	88 (0.7)
E: Treatment/intervention required – caused temporary harm	13 (0.1)
F: Initial/prolonged hospitalisation – caused temporary harm	4 (<0.1)
Not specified	4 (<0.1)

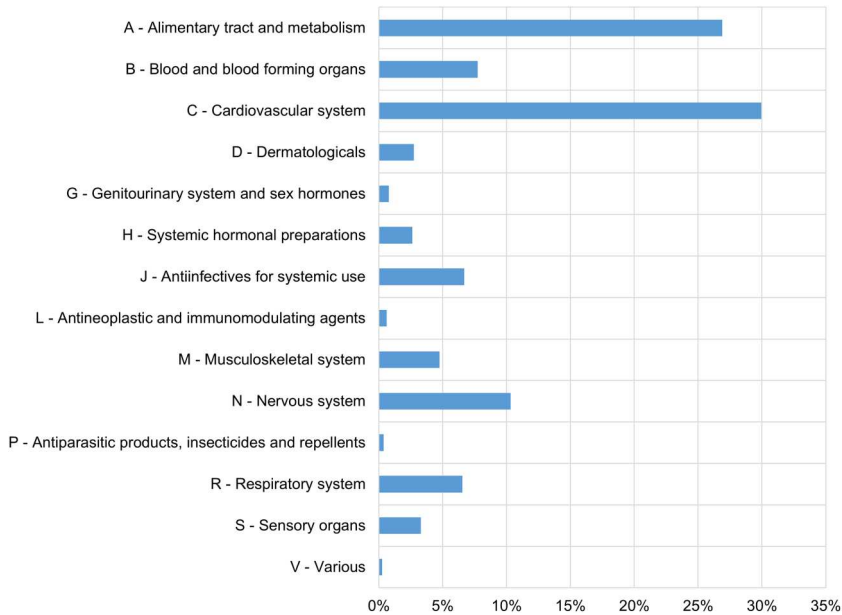


Figure 2. Therapeutic classes of drugs involved in medication errors due to transcribing errors. Percentages do not add up to 100% due to multiple drugs per report.

Table 3. Possible error causes and contributing factors to transcribing error.

	Total *n* (%)
Staff factors	
Inexperienced personnel	743 (6.2)
Inadequate knowledge	205 (1.7)
Distraction	4265 (35.4)
Medication related	
Sound alike medication	614 (5.1)
Look alike medication	149 (1.2)
Look alike packaging	68 (0.6)
Task and technology	
Failure to adhere to work procedure	1007 (8.4)
Use of abbreviations	58 (0.5)
Illegible prescriptions	333 (2.8)
Unavailable/inaccurate patient information	163 (1.4)
Wrong labelling/instruction on dispensing envelope or container	788 (6.5)
Work and environment	
Heavy workload	1856 (15.4)
Peak hour	4974 (41.3)
Stock arrangements/storage problem	43 (0.4)
Others	1702 (14.1)

Percentages do not add up to 100% due to multiple factors cited per report.

and systemic antibacterials were most frequently involved in the transcribing errors. Of note, insulins accounted for 35% of all drugs used in diabetes involved in errors.

Table 3 summarises possible causes and factors contributing to errors cited in the reports. Work environment was frequently cited, which included peak hours (41.3%) and heavy workload (15.4%). In addition, the occurrence of the errors was also attributed to staff distraction (35.4%). Of the factors related to task and technology, failure to adhere to work procedures was reported as the cause in 8.4% of cases. More importantly, illegible prescription was cited as the cause of error in 333 cases (2.8%) of transcribing errors.

Analysis stratified by facility setting is described in Table 4. Among ME cases reported by clinics, incorrect dose accounted for more than half of the total errors (54.0%), followed by incorrect frequency (16.8%) and incorrect drug (14.4%). Within hospitals, the most prevalent errors were incorrect dose (43.1%), incorrect frequency (23.5%), and incorrect drug (12.0%). No significant differences were observed in terms of the degree of harm from transcribing errors between hospitals and clinics. Analysis was also stratified by the health information status between full-based and pharmacy-based facilities. The types of errors reported were similar between the two types of facilities. Facilities with a full-based system had significantly higher proportions of errors that reached patients with a more severe degree of harm (category C and above), but the numbers of errors were small which potentially contributed to significant associations.

Table 4. Error type and outcome, stratified by (i) hospital and clinic, and (ii) health information system.

	Hospital, *n* (%)	Clinic, *n* (%)	*P*-value*	Full-based, *n* (%)	Pharmacy-based, *n* (%)	*P*-value*
Total errors, *N*	5706	6343		277	11,697	
Error type			<0.001			0.187
Dose error						
Incorrect dose	2462 (43.2)	3427 (54.0)		121 (43.7)	5739 (49.1)	
Incorrect frequency	1346 (23.6)	1064 (16.8)		54 (19.5)	2352 (20.1)	
Drug error						
Incorrect drug	685 (12.0)	923 (14.6)		44 (15.9)	1554 (13.3)	
Incorrect patient	396 (6.9)	369 (5.8)		17 (6.1)	738 (6.3)	
Dosage form error						
Incorrect dosage form	339 (5.9)	248 (3.9)		13 (4.7)	564 (4.8)	
Omission error						
Omission of drug	241 (4.2)	207 (3.3)		17 (6.1)	428 (3.7)	
Labelling error						
Incorrect instruction	58 (1.0)	6 (0.1)		1 (0.4)	63 (0.5)	
Other transcribing error	179 (3.1)	99 (1.6)		10 (3.6)	259 (2.2)	
Error outcome			0.152			<0.001
A: Potential error, events have potential to cause incident	38 (0.7)	43 (0.7)		1 (0.4)	80 (0.7)	
B: Actual error – did not reach patient (near miss)	5393 (94.5)	6034 (95.1)		242 (87.4)	11,132 (95.2)	
C: Actual error – caused no harm	208 (3.6)	224 (3.5)		26 (9.4)	393 (3.4)	
D: Additional monitoring required – caused no harm	55 (1.0)	33 (0.5)		7 (2.5)	75 (0.6)	
E: Treatment/ intervention required – caused temporary harm	8 (0.1)	5 (0.1)		1 (0.4)	10 (0.1)	
F: Initial/prolonged hospitalisation – caused temporary harm	2 (<0.1)	2 (<0.1)		–	3 (<0.1)	
Not specified	2 (<0.1)	2 (<0.1)		–	4 (<0.1)	

*Chi-square test for comparison between groups.

Factors associated with errors leading to patient harm were analysed by comparing characteristics between patients with MEs in categories A–D and those with categories E–F (Table 5). A significant association was observed for the status of health information system at the facilities and the timing of event. In a subgroup analysis, we described 534 MEs related to transcribing errors that reached patients (see Table S4). Half of the cases occurred in hospitals (50.6%) and 49.4% in clinics. Notably, 55.4% of the ME cases resulted in incorrect medication taken by patients. Dose error was the most prevalent (55.6%), followed by incorrect patient (17.6%) and incorrect drug (16.5%).

Table 5. Characteristic of transcribing errors stratified by degree of harm.

	A–D, *n* (%)	E–F, *n* (%)	*P*-value
Facility – type			0.343
Clinic	6334 (52.7)	7 (41.2)	
Hospital	5694 (47.3)	10 (58.8)	
Facility – status of health information system[a]			<0.001
Full-based	276 (2.3)	1 (5.9)	
Pharmacy-based	11,680 (97.1)	13 (76.5)	
Manual	59 (0.5)	2 (11.8)	
Time of event			0.043
Outside office hours	718 (6.0)	3 (17.7)	
Office hours	11,310 (64.0)	14 (82.3)	
Error type			0.172
Dose error	8286 (68.9)	10 (58.8)	
Drug error	2368 (19.7)	4 (23.5)	
Dosage form error	586 (4.9)	1 (5.9)	
Omission error	448 (3.7)	–	
Labelling error	64 (0.5)	–	
Other error	276 (2.3)	2 (11.8)	

*Univariate analysis of factors associated with errors that caused harm using Chi-square or Fisher exact test.

Although the majority of the events did not cause harm, 17 cases resulted in temporary harm and required intervention (3.1%).

Discussion

This study provides an overview of the frequency, reporting patterns, and characteristics of medication errors (MEs) related to the transcribing process in public health facilities in Malaysia. Transcribing errors were estimated at 3.5 events per 100,000 prescriptions, with a range of 1.13–5.30 per 100,000 prescriptions between 2018 and 2022.

Since the establishment of the Medication Error Reporting System (MERS) in Malaysia, the Ministry of Health (MOH) has consistently promoted the reporting of MEs through standardised and structured approaches. These efforts have included various outreach programs emphasising promotion, education, and training for healthcare professionals. The increasing trend in ME reports reflects a growing culture of voluntary reporting and a shift among healthcare professionals toward embracing ME reporting as an integral part of a robust healthcare system. This is crucial, as fostering a strong reporting culture can enhance medication processes and ultimately reduce the risk of harm to patients. Furthermore, ME reporting has become a key performance indicator (KPI) for pharmacists which further contributes to the growing trend, along with encouragement from the healthcare system to actively report such incidents. Previous study that used MERS data during 2009–2012 reported that 85% of all MEs occurred during prescribing process while 14% occurred during dispensing (Samsiah et al., 2016). Our current study provides an extension of this finding, by analysis of a more

recent data and also focusing on occurrence of transcribing errors. Comparing the results between the two study periods, the proportion of ME events from dispensing process during 2018–2022 was higher compared to 2009–2012 data (28% vs 14%). This could partly be attributed to increased number of ME reporting among the pharmacists over the years, as well as the shift towards computerisation of healthcare systems that may lead to opportunity for medication errors in other process. Moving forward, a comprehensive analysis of MEs that occurred at various stages of medication use process will be informative to further triangulate the ME scenarios in Malaysia and to design a more efficient intervention plan.

In Malaysia, the Pharmacy Information System (PhIS) has been implemented at Ministry of Health facilities in various stages over the past decades with the primary aim to streamline the medication order and dispense process, and pharmacy inventory (Ahmad Shanniza et al., 2022). However, many facilities which have yet to adopt electronic prescriptions or with partially digitalised systems still rely on manual systems, where doctors use paper prescriptions. During the process, prescription information is manually transcribed into the electronic pharmacy information system for medication dispensing. This manual data entry process is the potential point of errors. Our study showed that the proportion of transcribing errors was highest in facilities with pharmacy-based electronic health system (97.5%). This result was expected since facilities with pharmacy-based system are partially digitalised where prescriptions need to be transcribed by the pharmacy staff before being dispensed and administered to patients. Given the high number of prescriptions being transcribed daily, the probability of transcribing errors also increased. Previous studies have shown that electronic medication systems resulted in reduced medication error rates, particularly by eliminating the risk associated with unclear or incomplete orders from ambiguous handwriting (Jheeta & Franklin, 2017; Radley et al., 2013; Westbrook et al., 2012). Therefore, transcribing errors that were identified from our current study could potentially be reduced with the full implementation of PhIS and computerised medication order system. Nevertheless, excessive reliance on computerised order entry systems could introduce new types of errors stemming from human interaction with technology (Korb-Savoldelli et al., 2018), highlighting the need for proper training and safeguards. We also observed a small proportion of transcribing errors that occurred in facilities with full-based system, which warranted further assessment on whether the errors occurred due to system breakdown or other factors. We extracted data on transcribing errors based on ME reports classified as 'data entry system error'. Hence, facilities with the manual system are less likely to report transcribing errors since they do not have data entry process in medication supply workflow with the exception of several facilities that incorporated a stand-alone record keeping system.

The most common types of transcribing errors identified in this study were incorrect dose (48.6%), incorrect frequency (19.9%), and incorrect drug (13.3%). Knudsen et al. explained these errors by highlighting three main problem areas in their study: difficulties in reading or decoding prescriptions accurately, selecting the correct option in the computerised system, and ensuring precise manual data entry (Knudsen et al., 2007). They also found that one of the key root causes of frequently reported transcription errors related to strength and dosage was the presence of 'traps' – factors that create confusion or misinterpretation. In our present study, medications that were most frequently identified with transcribing errors are those commonly prescribed in healthcare facilities and often prescribed with multiple medications for chronic diseases, such as antidiabetics, lipid-modifying drugs, and antihypertensives. Antidiabetics include insulin, which is regarded as a high-risk drug due to the different types of formulations available and complex dosing regimen that make it prone to medication error (McKay et al., 2022; Medication Safety in High-risk Situations, 2019). Previous reports have shown that transcribing a handwritten prescription with insulin can induce errors as it can be easily misread (e.g. 4U of insulin misinterpreted as 44 units) (National Diabetes Inpatient Audit (NaDIA), 2013; Parmar, 2005; Pennsylvania Patient Safety Authority, 2010). Similarly for other drugs, in which different formulations or strengths that are available often result in errors during transcribing, as reflected in the high frequency of errors regarding incorrect dose in our findings.

We observed that majority of the transcribing errors did not reach patients as these errors were detected before dispensing. This is encouraging as it reflects the practice of good dispensing at these healthcare facilities which involves checking of the prescription and filled medicines to ensure that the correct medicine is supplied to the correct patient. The checking process are implemented in the pharmacy department as part of the routine practice in pharmacy service (Pharmaceutical Services Division. Ministry of Health Malaysia, 2016; World Health Organization, 2011). However, if such errors can be avoided altogether, it can further reduce the workload burden as our study also reported 534 cases of transcribing errors that reached patients. We also found that most of the transcribing errors occurred during office hours when the workload is at its peak, with a high volume of prescriptions to process. Another frequently cited causes of errors are staff-related factors, such as distraction. This may be attributed to the need for staff to multitask during these busy periods, including handling multiple prescriptions, clarifying prescriptions with prescribers or patients, and providing patient counselling simultaneously. Such demands can compromise staff concentration, ultimately increasing the likelihood of medication error.

There are several limitations to our study. First, the type and number of reports collected are highly dependent on the accuracy and completeness of the submitted reports. Additionally, there is limited information available on

MEs from private healthcare facilities and university hospitals; therefore, we only included data from Ministry of Health Malaysia institutions. Currently, ME reports from the private sector and university hospitals account for less than 5% of reports within the MERS database. The exclusion of the data is unlikely to significantly impact the results of our analysis. Nevertheless, to our knowledge, this is the first study to describe the occurrence of transcribing errors at healthcare facilities at the national level in Malaysia and using most recent data from the MERS database. Besides the characteristics and outcomes of the transcribing errors, we also described the frequencies of transcribing errors by the type of electronic health systems implemented at the facilities. As more hospitals and clinics adopt and transition to electronic systems, future research could focus on evaluating the impact of these systems on the prevalence and nature of MEs. This would provide a more comprehensive understanding of how digitalisation influences medication safety.

Conclusion

This study described the prevalence and types of transcribing errors in public healthcare facilities in Malaysia from medication errors reported in the Medication Error Reporting System (MERS). We observed an increase in voluntary reporting over the years, reflecting a growing culture of reporting among healthcare professionals. While transcribing errors accounted for less than 5% of all medication errors reported with a prevalence of 35 events for every 1 million prescriptions processed, these errors can potentially be prevented with appropriate intervention. Our findings offer valuable insights into medication errors caused by transcribing processes in Malaysia, which can inform efforts to improve pharmacy practices and ultimately, reduce harm to patients.

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Authors' contributions

FK, NT, EVL, and NAR devised the study and refined the main conceptual ideas. FK, ACA, SCC, NT, and NI were involved in data acquisition and processing. NAR analysed the data with input and feedback from the team. NAR and EVL wrote the initial draft of the manuscript. All authors provided critical input, edited the manuscript, and approved the final version.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Ethical approval

This study was registered with the National Medical Research Register and approved by the Medical Research and Ethics Committee, Ministry of Health Malaysia (NMRR ID-24-00304-OBO).

Availability of data and materials

The data that support the findings of this study are available upon reasonable request. Access to MERS database needs to be requested to the organisation owning the data.

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